

## Task 01 : Customer Conversation

| Client Issues                                                                                                                                                                                                                                                                                            | AWS Solution Builder Response                                                                                                                                                                                                                                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>The promotional event we held last week increased our web traffic and overloaded our web application. All requests were being timed out.</p> <p>Worse than that, our Amazon EC2 instances were also hosted in an Availability Zone that was affected by severe storms. We were offline for hours.</p> | <p>you can use AWS to create an Auto Scaling group across multiple Availability Zones so that your resources aren't isolated.</p> <p>you can create a load balancer so that network traffic is distributed equally across your servers.</p>                       |
| <p>Okay, so if I spread instances out over multiple AZs, will that make us completely safe from failures?</p>                                                                                                                                                                                            | <p>Well, no system is 100 percent failure-proof, but Availability Zones reside in geographically distinct locations, and a singular event is unlikely to affect multiple AZs simultaneously.</p>                                                                  |
| <p>how can we verify that our application is down in a different Availability Zone?</p>                                                                                                                                                                                                                  | <p>After you spread your EC2 instances over multiple AZs, you can attach a load balancer to your Auto Scaling group for high availability and elasticity.</p> <p>You can also monitor EC2 instance health by setting up a health check on your load balancer.</p> |

## Task 2 : Questions based on steps

1. What type of load balancer is used here and why?

Application Load Balancer

health checks to provide high availability.

Application Load Balancer helps to Active and passive node configuration both so In the Application Load balancer helps to high available on servers

Application Load Balancer performs health checks on registered targets (EC2 instances, containers, IPs) in a target group. It periodically sends requests (HTTP/HTTPS) to each target and marks them as healthy or unhealthy based on the response. Traffic is only routed to targets marked "healthy."

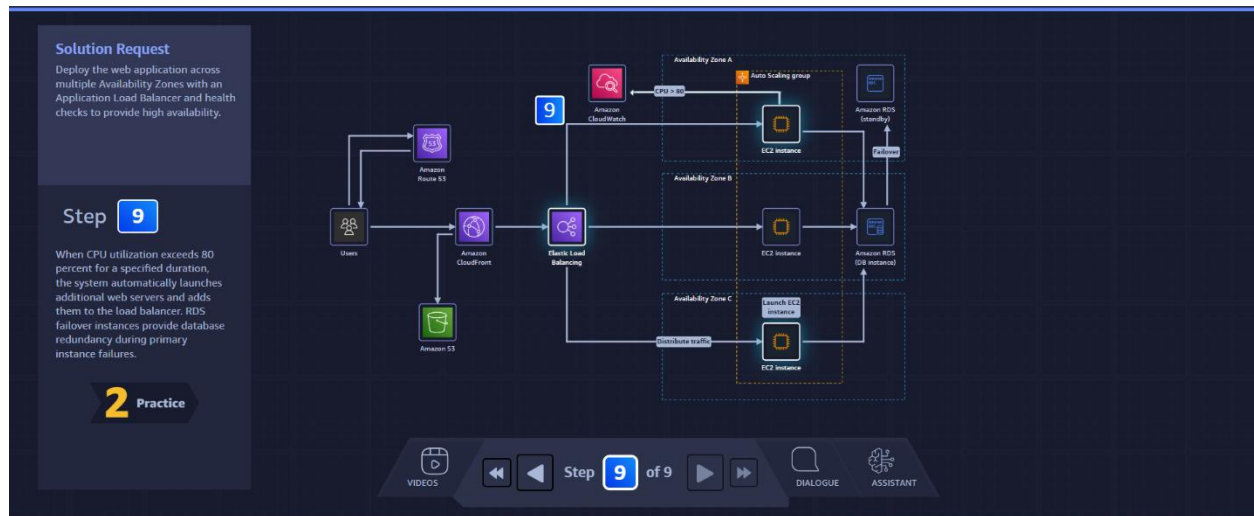
2. What are the desired capacity and scaling limits you configured?

- in the Practice session I have Configured 2 group and configured as capacity of 2 and scaling limit also was 2
- in the DIY session I have Configured 2 group and configured as capacity of 3 and scaling limit also was 3

3. What happens when you span your Auto Scaling group across multiple Availability Zones?

Auto scaling mainly do the keep balancing of Instances and availability zones . When it would be high availability though auto scaling .

### Task 3: Provide your proof of Activity Completion



EC2 > Auto Scaling groups

**Auto Scaling groups (1/1) Info**  
Last updated less than a minute ago

[Launch configurations](#) [Launch templates](#) [Actions](#) [Create Auto Scaling group](#)

Search your Auto Scaling groups

| Name                                   | Status              | Launch template/configuration                          | Instances |
|----------------------------------------|---------------------|--------------------------------------------------------|-----------|
| <a href="#">TravelAgencyWebServers</a> | At desired capacity | <a href="#">TravelAgencyTemplate-4c10e500</a>   Versio | 1         |

**Auto Scaling group: TravelAgencyWebServers**

**TravelAgencyWebServers Capacity overview** [Edit](#)

[arn:aws:autoscaling:us-east-1:515966494430:autoScalingGroup:dde965c2-eb21-46ed-9e6b-fede236ce6fc:autoScalingGroupName/TravelAgencyWebServers](#)

EC2 > Target groups > Create target group

**Step 1 Create target group**  
Step 2 - recommended Register targets  
Step 3 Review and create

**Create target group**  
A target group can be made up of one or more targets. Your load balancer routes requests to the targets in a target group and performs health checks on the targets.

**Settings - immutable**  
Choose a target type and the load balancer and listener will route traffic to your target. These settings can't be modified after target group creation.

**Target type**  
Indicate what resource type you want to target. Only the selected resource type can be registered to this target group.

☒ **Instances**  
Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or ECS services for automatic management.  
Suitable for: [ALB](#) [NLB](#) [GWLB](#)

☐ **IP addresses**  
Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same instance. Supports IPv6 targets.  
Suitable for: [ALB](#) [NLB](#) [GWLB](#)

☐ **Lambda function**  
Supports load balancing to a single Lambda function. ALB required as traffic

☐ **Application Load Balancer**  
Allows use of static IP addresses and RequestId with an Application Load

group for the rule action.

## How Application Load Balancers work

### Basic configuration

#### Load balancer name

Name must be unique within your AWS account and can't be changed after the load balancer is created.

TravelAgencyServers-ALB

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

#### Scheme

Scheme can't be changed after the load balancer is created.

##### Internet-facing

- Serves internet-facing traffic.
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.

##### Internal

- Serves internal traffic.
- Has private IP addresses.
- DNS name resolves to private IPs.
- Compatible with the IPv4 and Dualstack IP address types.

## Create security group

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

### Basic details

#### Security group name

TravelAgencyLoadBalancer

Name cannot be edited after creation.

#### Description

Allow Access to the Travel Agency load balancer From the internet

#### VPC

vpc-09d12939844382b9b (lab/TravelAgencyVpc)

- Elastic Block Store
- Volumes
- Snapshots
- Lifecycle Manager

### Network & Security

#### Security Groups

Elastic IPs

Placement Groups

Key Pairs

Network Interfaces

#### Load Balancing

Load Balancers

Target Groups

Trust Stores

#### Auto Scaling

Auto Scaling Groups

Settings

Security group (sg-07a8394778c9b3756 | TravelAgencyLoadBalancer) was created successfully

Details

### Security Groups (1/5)

Find security groups by attribute or tag

| Name | Security group ID    | Name      | VPC ID                |
|------|----------------------|-----------|-----------------------|
| -    | sg-07a8394778c9b3756 | dBalancer | vpc-09d12939844382b9b |
| -    | sg-005f61202a386412f | bServer   | vpc-09d12939844382b9b |

#### sg-005f61202a386412f - TravelAgencyWebServer

Details Inbound rules Outbound rules

### Details

| Security group name | Security group ID | Description | VPC ID |
|---------------------|-------------------|-------------|--------|
|---------------------|-------------------|-------------|--------|

EC2 > Load balancers > TravelAgencyServers-ALB

▼ Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

▼ Network & Security

Security Groups

Elastic IPs

Placement Groups

Key Pairs

Network Interfaces

▼ Load Balancing

Load Balancers

Target Groups

Trust Stores

▼ Auto Scaling

Auto Scaling Groups

Settings

Monitor how much of your load balancer's available capacity was used to meet traffic demand. Available in both the **Capacity** and **Monitoring** tabs.

TravelAgencyServers-ALB

⌂ Actions ▼

▼ Details

Load balancer type

Application

Status

✔ Active

VPC

[vpc-09d12939844382b9b](#) ⓘ

Load balancer IP address type

IPv4

Scheme

Internet-facing

Hosted zone

Z35XDOTRQ7X7K

Availability Zones

[subnet-0dcccfa242817523e](#) ⓘ us-east-1c (use1-az4)

[subnet-0e41346b4a6cb2454](#) ⓘ us-east-1a (use1-az1)

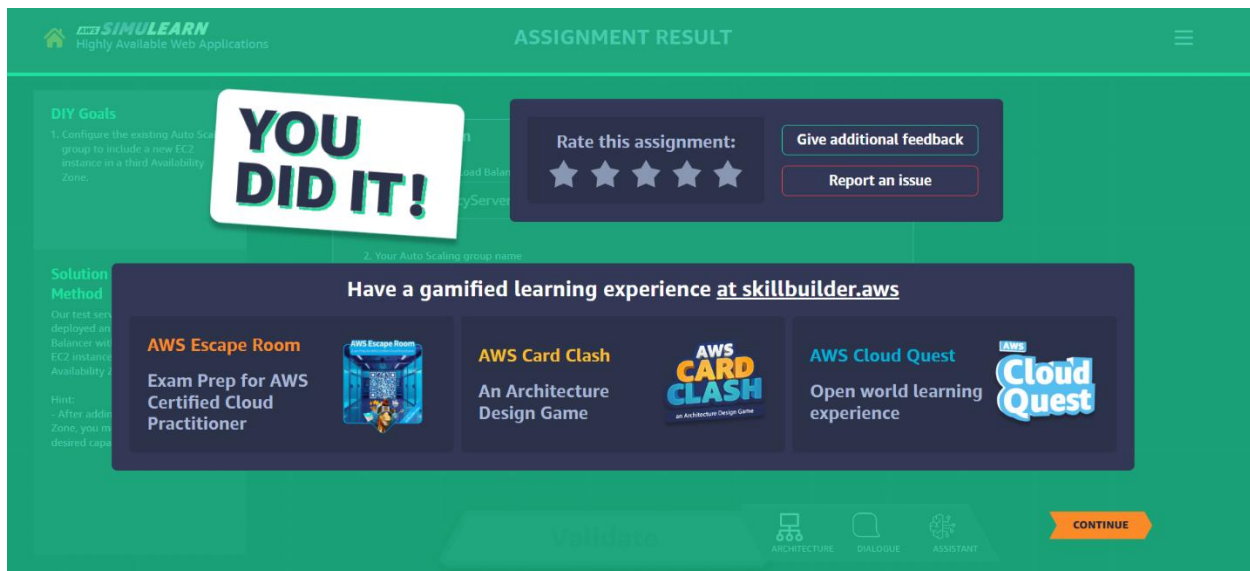
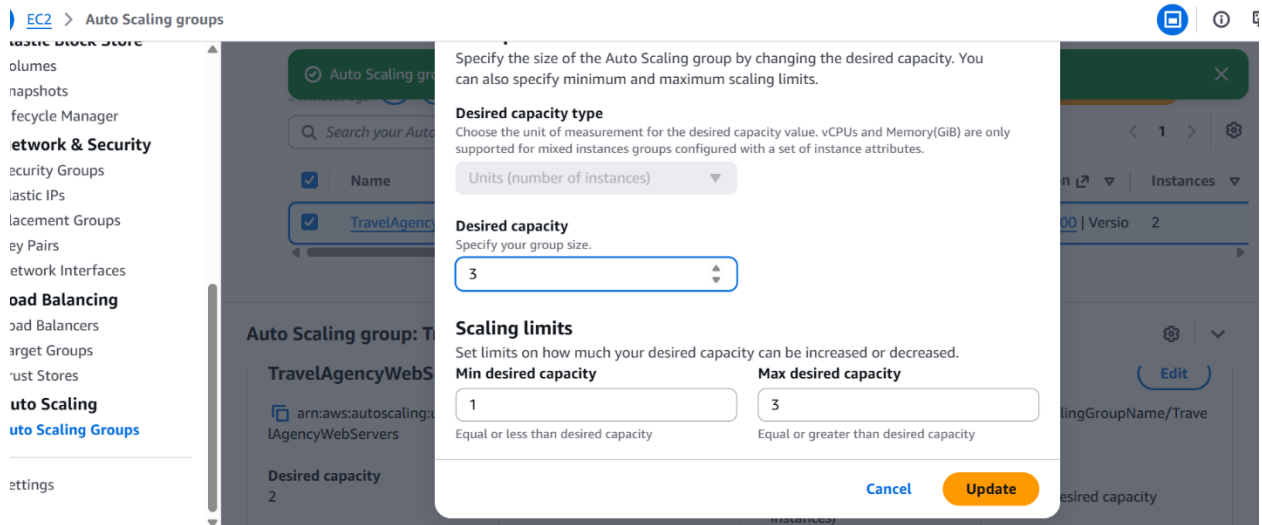
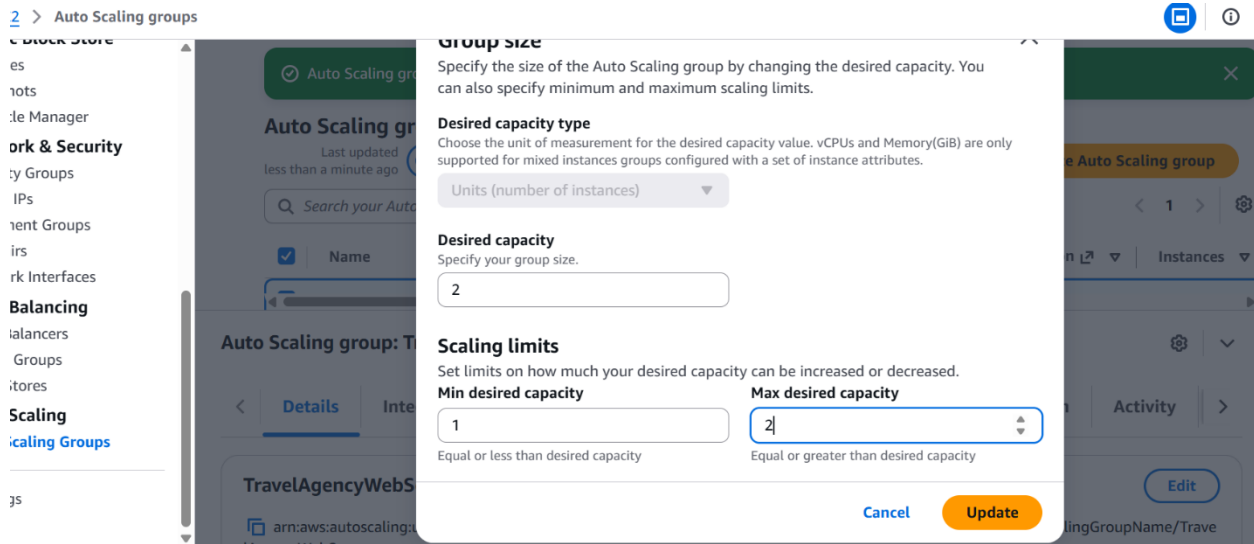
[subnet-03f7c639ed10fc514](#) ⓘ us-east-1b (use1-az2)

Date created

August 22, 2026, 12:33 (UTC+05:30)

A landing page for Unicorn Travel Agency. The background is a dark, moody image of a cloudy sky at sunset or sunrise, with purple and orange hues. The text "UNICORN TRAVEL AGENCY" is centered in a large, bold, sans-serif font. "UNICORN" is in a reddish-brown color, and "TRAVEL AGENCY" is in a lighter, greyish-brown color. In the top left corner, there is a "Sign In" link. In the top right corner, there are "About" and "Help" links.

Instance i-02f23a4bd96b7f780 is healthy



**AWS Training & Certification**  
Completion Certificate

# AWS SimuLearn: Highly Available Web Applications

Completed: August 22, 2026

Awarded to  
Sadumina Bagya Rathnayaka

